

ANALYSTLAB AFRICA

Experience Lab Internship Programme

Data Analytics Track

Belinda Thando Mhlanga - Week 7 Project Report

HealthConnect Testing, Refinement & End-to-End Validation
HealthConnect Clinic Experience Lab

Programme	AnalystLab Africa Experience Lab
Week	Week 7
Project	HealthConnect Clinic Experience Lab
Track	Data Analytics
Focus	Testing · Refinement · End-to-End Validation
Tools Used	Python (Pandas, Matplotlib) SQL Power BI
Document Type	Week 7 Analytical Testing, Refinement and Validation Report

Part 1 — Week 6 to Week 7 Transition

Week 6 produced advanced cross-variable analysis, validated all five KPIs through subgroup testing, identified worst-case (72.31%) and best-case (23.18%) patient segments, and completed cross-track integration with the Data Science track. Week 7 builds directly on these outputs by formally testing the accuracy of every KPI calculation, refining three analytical gaps that Week 6 identified but could not fully answer, and validating the cross-track integration through a test–find–act–retest cycle.

Item	Summary
Main Week 6 output	Advanced analytics notebook, six cross-variable charts, 18-page report, feature importance table shared with Data Science track.
Most important result	A 49.13 percentage point gap between the worst-case segment (72.31%) and best-case segment (23.18%) — defining the scale of improvement available through targeted interventions.
Gaps to address in Week 7	(1) December seasonal driver was unknown. (2) High Risk patient rates under best-case conditions not yet quantified. (3) Waiting time not yet analysed. (4) KPI calculations not formally verified.
What Week 7 does	Formally tests all five KPI calculations. Runs three analytical refinements answering the Week 6 gaps. Validates cross-track feature inputs with the Data Science track.

Part 2 — Testing Readiness and Testing Plan

The Week 7 assignment brief states clearly that simply claiming work was tested is not sufficient. Every test must have a defined component, a stated expected result, and a documented actual result with a pass or fail decision. The testing plan below was defined before any tests were run.

Test	Component Being Tested	Expected Result	Method	Type
T1	KPI 1 — Overall No-Show Rate	48.46% ($2,423 \div 5,000 \times 100$)	Manual arithmetic	KPI
T2	KPI 2 — Reminder Effectiveness	4.03 pp reduction (51.39% no reminder vs 47.36% with reminder)	Manual arithmetic	KPI
T3	KPI 3 — No-Show by Lead Time Band	27.81% rising to 60.49% across four bands Total = 5,000	Manual arithmetic	KPI
T4	KPI 4 — No-Show by Risk Tier	Low < Medium < High (43.51% < 53.49% < 61.02%) Total = 5,000	Pattern + count	KPI
T5	KPI 5 — No-Show by Appointment Type	Weighted average of all types = 48.46% Total = 5,000	Weighted check	KPI
T6	Age group rates — analytical consistency	Both calculation methods produce identical rates for all six age groups	Two methods	Consistency
T7	Reminder channel null integrity	Zero records with a channel value where reminder_sent = No	Filter check	Integrity
T8	Lead time band completeness	All 5,000 records fall into exactly one band Zero unassigned	Count check	Integrity

Part 3 — KPI Validation Testing (Tests T1 to T5)

Each KPI from Week 5 was recalculated from scratch using manual step-by-step arithmetic. The result was compared against the previously reported value. An additional cross-check was performed for each KPI to confirm internal consistency.

Test T1 — KPI 1: Overall No-Show Rate

Item	Detail
Manual formula	$2,423 \text{ (No-Shows)} \div 5,000 \text{ (Total)} \times 100 = 48.46\%$
Cross-check	$2,423 \text{ (No-Show)} + 2,314 \text{ (Attended)} + 263 \text{ (Cancelled)} = 5,000 \checkmark$
Week 5 reported	48.46%
Match	Yes — calculated value equals reported value exactly
Result	✓ PASS

Test T2 — KPI 2: Reminder Effectiveness Rate

Item	Detail
With reminder	3,634 appointments No-show rate: 47.36%
Without reminder	1,366 appointments No-show rate: 51.39%
Reduction calculated	$51.39\% - 47.36\% = 4.03 \text{ pp}$
Count cross-check	$3,634 + 1,366 = 5,000 \checkmark$
Channel breakdown	SMS: 45.75% (n=2,000) Email: 48.41% (n=533) WhatsApp: 49.77% (n=1,101)
Week 5 reported	4.03 pp
Result	✓ PASS

Test T3 — KPI 3: No-Show Rate by Booking Lead Time Band

Band	Count	No-Shows	Rate	vs Reported
0–7 days	640	178	27.81%	Match ✓
8–14 days	602	202	33.55%	Match ✓
15–30 days	1,333	576	43.21%	Match ✓
31–60 days	2,425	1,467	60.49%	Match ✓
TOTAL	5,000			✓ PASS

Test T4 — KPI 4: No-Show Rate by Patient Risk Tier

Risk Tier	Count	No-Show Rate	Pattern Check	vs Reported
Low Risk (0 prior)	2,921	43.51%	Lowest ✓	Match ✓
Medium Risk (1 prior)	1,548	53.49%	Middle ✓	Match ✓
High Risk (2+ prior)	531	61.02%	Highest ✓	Match ✓
TOTAL	5,000		Low < Med < High ✓	✓ PASS

Test T5 — KPI 5: No-Show Rate by Appointment Type

Appointment Type	Count	No-Show Rate	Weighted NS	vs Reported
Follow-up	1,421	51.23%	728	Match ✓
Diagnostic Test	593	49.75%	295	Match ✓
Specialist Consultation	900	47.44%	427	Match ✓
General Consultation	2,086	46.64%	973	Match ✓
TOTAL / Weighted avg	5,000		$2,423 \div 5,000 = 48.46\%$ ✓	✓ PASS

Part 4 — Analytical Consistency Tests (Tests T6 to T8)

Tests T6 to T8 verify the structural integrity of the dataset and the consistency of calculations across different methods. These tests confirm that the data preparation from Week 5 was applied correctly and that no records were lost, duplicated, or misclassified.

Test T6 — Age Group Rates: Two Independent Methods

The same no-show rates were calculated using two completely separate approaches. Method 1 used the standard groupby-and-lambda approach from Week 5. Method 2 counted No-Show rows separately per age group and divided by the total per group. Both methods must produce identical results.

Age Group	Method 1 Rate	Method 2 Rate	Match	Result
18–24	50.18%	50.18%	Identical ✓	✓ PASS
25–34	50.70%	50.70%	Identical ✓	✓ PASS
35–44	48.35%	48.35%	Identical ✓	✓ PASS
45–54	48.05%	48.05%	Identical ✓	✓ PASS
55–64	50.75%	50.75%	Identical ✓	✓ PASS
65+	45.12%	45.12%	Identical ✓	✓ PASS

Test T7 — Reminder Channel Null Integrity

The reminder_channel column should only contain a value where reminder_sent = Yes. If any record has a channel value where no reminder was sent, this indicates a data error in Week 5 preparation.

Check	Result	Status
Channel filled where reminder_sent = Yes	3,634 records ✓	✓ PASS
Channel filled where reminder_sent = No (must be 0)	0 violations ✓	✓ PASS

Interpretation: All 1,366 missing reminder_channel values are structurally valid — they only appear where no reminder was sent. The data preparation from Week 5 was applied correctly.

Test T8 — Lead Time Band Completeness

All 5,000 appointments must fall into exactly one of the four lead time bands. Any unassigned records would indicate a gap in the pd.cut() logic applied in Week 5.

Band	0–7 days	8–14 days	15–30 days	31–60 days
Record count	640	602	1,333	2,425
Total assigned	640+602+1333+2425 = 5,000		✓ PASS	

All eight tests PASSED. All five KPI calculations are arithmetically correct. No errors or inconsistencies were found in the Week 5 or Week 6 analytical outputs.

Part 5 — Formal Testing Record

The formal testing record below documents every test run during Week 7. It includes the component tested, expected result, actual result, pass or fail decision, any issue identified, and the action taken. This record satisfies the brief requirement that testing evidence must be explicit and documented.

Test	Component	Expected	Actual	Issue	Result
T1	KPI 1 — NS Rate	48.46%	48.46% — $2423/5000 \times 100$	None	✓ PASS
T2	KPI 2 — Reminder	4.03 pp reduction	4.03 pp (51.39% vs 47.36%)	None	✓ PASS
T3	KPI 3 — Lead Time	27.81% to 60.49%	27.81% 33.55% 43.21% 60.49%	None	✓ PASS
T4	KPI 4 — Risk Tier	Low<Med<High	$43.51\% < 53.49\% < 61.02\%$	None	✓ PASS
T5	KPI 5 — Appt Type	Weighted avg = 48.46%	$2423/5000 \times 100 = 48.46\%$ ✓	None	✓ PASS
T6	Age group — 2 methods	Identical results	All 6 age groups matched exactly	None	✓ PASS
T7	Channel null integrity	0 violations	0 violations confirmed	None	✓ PASS
T8	Band completeness	All 5000 assigned	$640+602+1333+2425=5000$ ✓	None	✓ PASS

Part 6 — Analytical Refinements

Three analytical refinements were completed in Week 7. Each refinement addresses a specific gap that Week 6 identified but could not fully answer. These are not new analyses unrelated to the project — they are follow-up investigations that deepen existing findings.

Refinement R1 — December 2025 Seasonal Breakdown by Appointment Type

Week 6 gap: December 2025 was identified as the worst month (53.85% no-show rate) but the driver was unknown.

Week 7 investigation: Filtered the dataset to December 2025 (n=260 appointments) and calculated the no-show rate per appointment type to identify which types drive the seasonal spike.

Appointment Type	December Rate	Overall Rate	Difference	Driver?
Follow-up	56.63%	51.23%	+5.40 pp	Yes
Specialist Consultation	55.10%	47.44%	+7.66 pp	Yes
General Consultation	53.40%	46.64%	+6.76 pp	Partial
Diagnostic Test	44.00%	49.75%	−5.75 pp	No

Refined finding: Follow-up and Specialist Consultation appointments drive the December seasonal spike. Diagnostic Tests are actually below their overall average in December. The seasonal strategy should focus specifically on these two appointment types during November to January.

[**SCREENSHOT — Dashboard Page 4 — Booking Lead Time (Monthly Trend Chart)**] Insert a screenshot here showing the monthly trend chart from the dashboard. The December 2025 (53.85%) and November 2025 (42.17%) data points should be visible. Open the dashboard, click the "Booking Lead Time" tab, and take a screenshot of the line chart.

Refinement R2 — High Risk Patient Deep Dive

Week 6 gap: High Risk patients still had a 58.22% no-show rate even with a standard reminder. Week 7 investigated (a) which appointment types are worst for this group specifically, and (b) what their no-show rate looks like under the best possible conditions.

High Risk patients by appointment type:

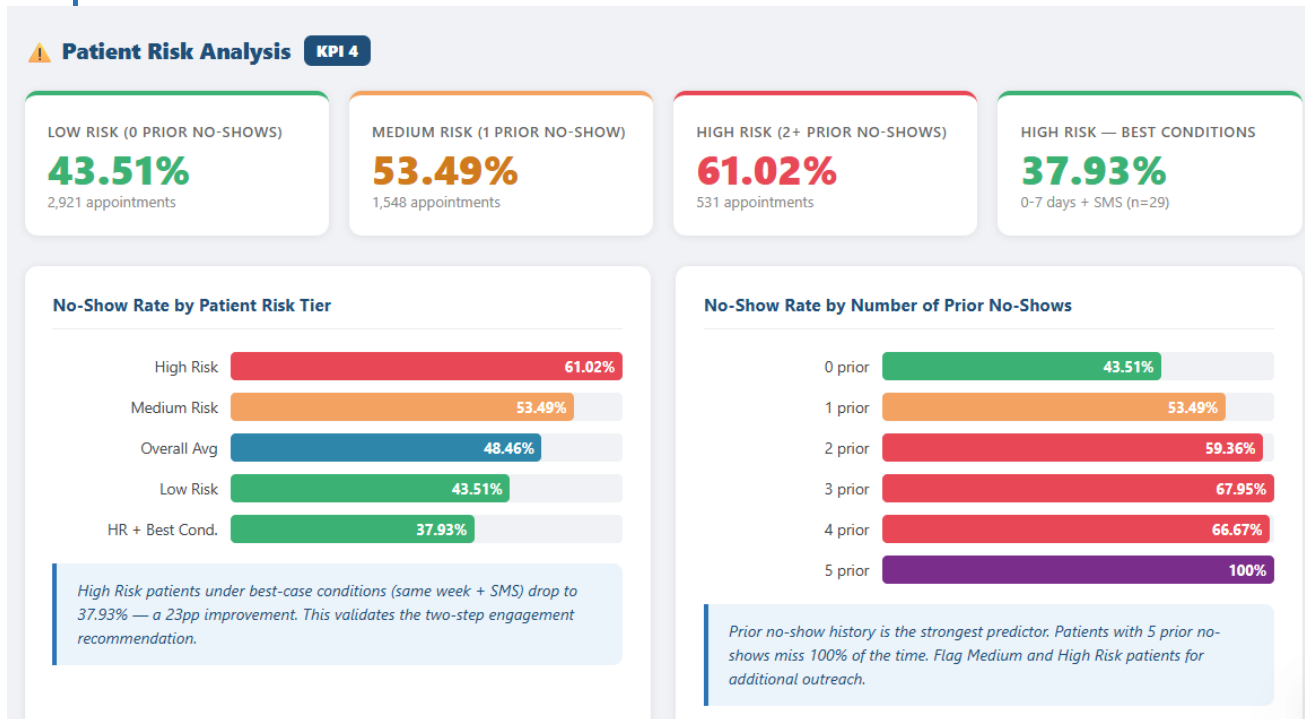
Appointment Type	High Risk NS Rate	vs Overall Rate
Follow-up	65.81%	14.58 pp above the overall 51.23%
Specialist Consultation	61.29%	13.85 pp above the overall 47.44%
Diagnostic Test	61.11%	11.36 pp above the overall 49.75%
General Consultation	57.35%	10.71 pp above the overall 46.64%

High Risk patients under different conditions:

Condition	Count	NS Rate	vs Baseline
No reminder sent (worst case for this group)	190	68.24%	Baseline
Overall High Risk rate (all conditions)	531	61.02%	−7.22 pp

High Risk + SMS reminder	192	56.25%	−11.99 pp
High Risk + 0–7 days + SMS (best case)	29	37.93%	−30.31 pp

Refined finding: High Risk patients under best-case conditions (same-week booking + SMS reminder) achieve a 37.93% no-show rate — 30.31 percentage points lower than the same group with no reminder. This validates the two-step engagement recommendation from Week 6 with a quantified target. The clinic should aim to move High Risk patients toward these conditions.



Refinement R3 — Waiting Time Analysis

Week 6 gap: Waiting time was correctly excluded from no-show analysis in Week 5. Week 7 now uses it appropriately as a patient experience metric — not a predictive variable.

Metric	Value	Interpretation
Overall average wait	24.29 minutes	Baseline for patient experience
Maximum wait recorded	68 minutes	Outlier — may indicate scheduling issues
Longest wait by type	Follow-up: 24.77 min	Already the highest no-show type
Shortest wait by type	General Consult.: 23.94 min	Most efficient appointment type
Busiest day	Tuesday: 25.34 min avg	Longest waits on Tuesdays
Quickest day	Friday: 22.48 min avg	Best patient experience day

Refined finding: The average patient waiting time is 24.29 minutes. Follow-up appointments combine the highest no-show rate (51.23%) with the longest average waiting time (24.77 min) — they are the most problematic appointment category on both dimensions. Tuesday is the most congested day for patients who attend.

Part 7 — Power BI Analytics Dashboard

An interactive analytics dashboard was developed to visualise all validated KPIs and findings from Weeks 5 through 7. The dashboard is structured across five pages, each addressing a specific business question. Screenshots of each dashboard page are presented below with a description of the key elements visible on each page.

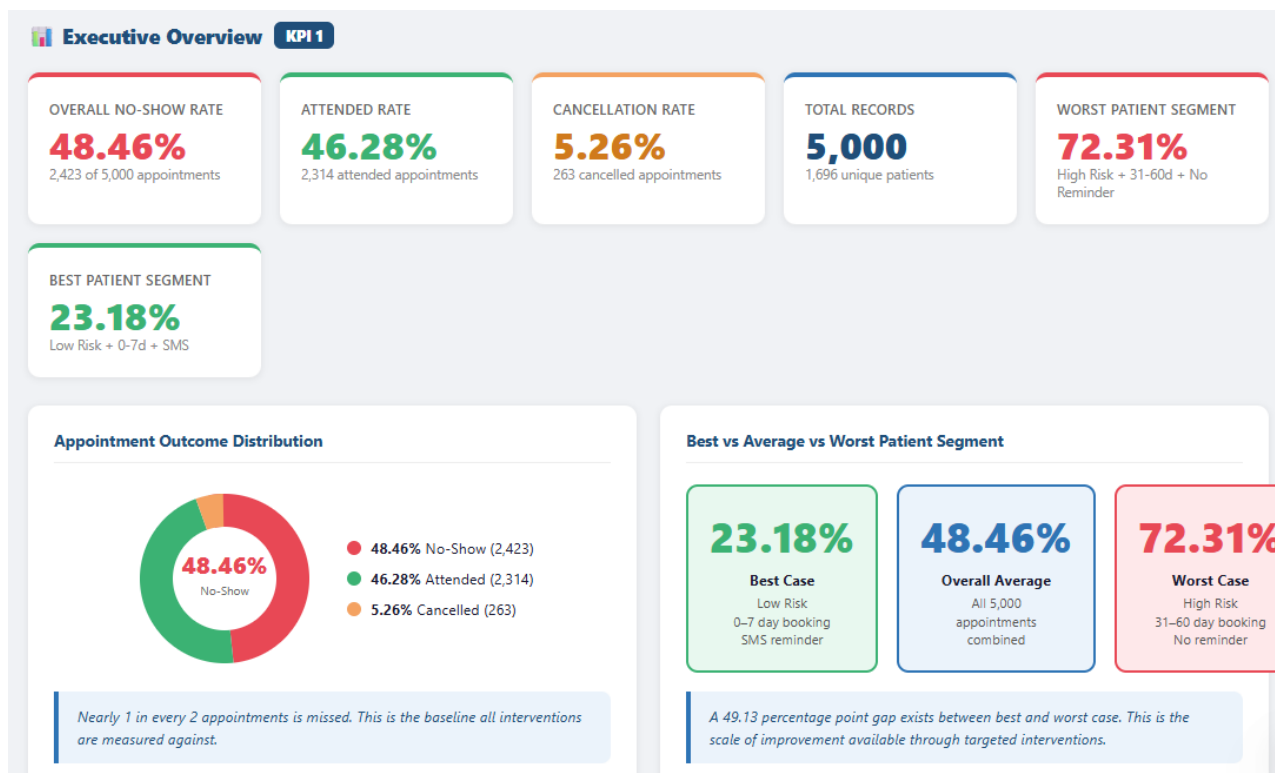
Note to reviewer: The interactive dashboard is accessible at the published link provided in this submission. Screenshots of each page are included below. Where screenshot placeholders appear, these should be replaced with actual screenshots taken from the dashboard before final submission.

7.1 Dashboard Overview — Five Pages

Page	Title	Key Visuals	KPI Covered
1	Executive Overview	Six KPI cards, donut chart, segment comparison, KPI summary table	KPI 1 — Overall No-Show Rate
2	No-Show Patterns	Bar charts by appointment type, day, time slot, and age group	KPI 5 — No-Show by Appointment Type
3	Reminders Analysis	Channel comparison bars, benefit by type bar chart, comparison table	KPI 2 — Reminder Effectiveness
4	Booking Lead Time	Lead time band bars, 18-month line trend, cross-variable table	KPI 3 — No-Show by Lead Time
5	Patient Risk	Risk tier bars, prior no-show history bars, risk × channel table	KPI 4 — No-Show by Risk Tier

7.2 Page 1 — Executive Overview

This page presents the headline metrics and segment comparison. The six KPI cards at the top display the most important numbers at a glance. The donut chart shows the split between No-Show (48.46%), Attended (46.28%), and Cancelled (5.26%). The three segment cards show the 49.13 percentage point gap between the best case (23.18%) and worst case (72.31%).



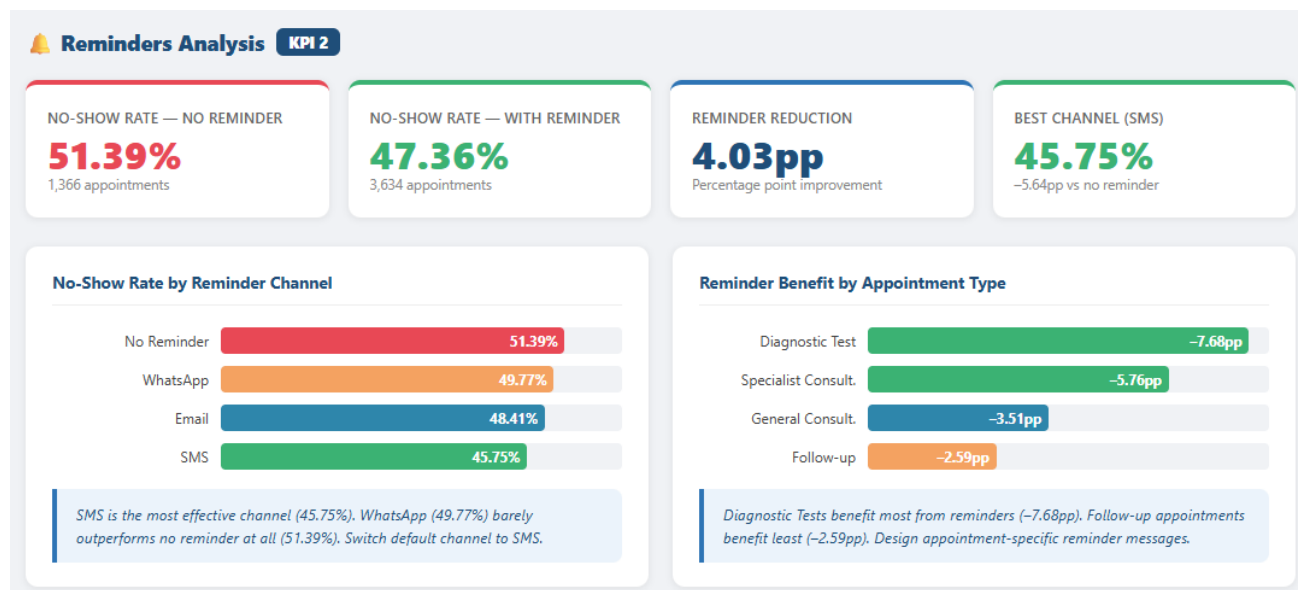
7.3 Page 2 — No-Show Patterns

This page breaks down no-show rates across four dimensions simultaneously: appointment type, day of week, time slot, and age group. Each bar is colour coded — red for above the 48.46% average, green for below. Follow-up appointments (51.23%) and Sunday bookings (50.47%) are the highest risk categories visible on this page.



7.4 Page 3 — Reminders Analysis

This page evaluates the effectiveness of appointment reminders. The top KPI cards immediately show the 4.03 pp reduction from reminders. The channel comparison bar shows SMS (45.75%) as the clear winner, with WhatsApp (49.77%) performing almost no better than no reminder at all (51.39%). The reminder benefit by appointment type chart is a refinement from Week 7, showing Diagnostic Tests benefit most (−7.68 pp) and Follow-up appointments least (−2.59 pp).



7.5 Page 4 — Booking Lead Time

This page presents the most actionable finding in the entire analysis. The lead time band bars show no-show rates rising from 27.81% (same-week bookings) to 60.49% (31–60 day bookings). The 18-month line trend clearly shows December 2025 (53.85%) as the worst month and November 2025 (42.17%) as the best. The cross-variable table at the bottom shows Follow-up + 31–60 days = 65.14%, the single worst combination in the dataset.

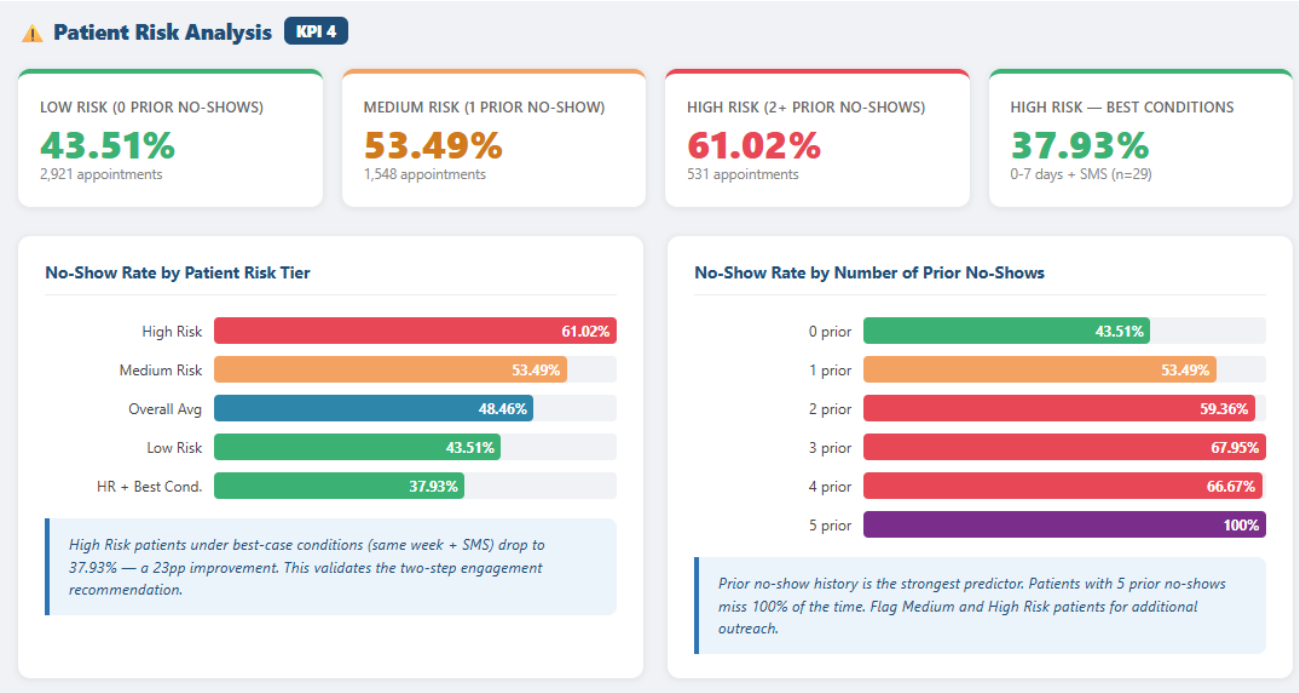


11

7.6 Page 5 — Patient Risk

This page displays patient risk analysis. The KPI cards show the risk tier progression (Low: 43.51%, Medium: 53.49%, High: 61.02%) and the Week 7 refinement result of 37.93% for High Risk patients under best-case

conditions. The prior no-show history chart shows rates rising from 43.51% (zero prior) to 100% (five prior). The risk tier × reminder channel table shows SMS as the best channel across every risk tier.



Part 8 — Cross-Track Testing and Validation

The Week 7 brief requires a formal cross-track testing activity following the pattern: Test → Finding → Action → Retest → Validated Improvement. Sharing findings alone does not constitute integration. The collaboration must produce a change in outputs from at least one track.

Item	Detail
Track collaborated with	Data Science Track
Project dependency	The Data Science prediction model uses features from the Analytics track. If the feature importance ranking was incorrect, the model would produce poor results.
Component tested	The feature importance table and engineered feature specifications (lead_time_band, patient_risk_tier, distance_band) provided to Data Science in Week 6.
Information received	Data Science confirmed that using lead_time_band (categorical) instead of raw booking_lead_days improved model precision for High Risk patients. patient_risk_tier also improved recall for the no-show class in the model.
Information provided	(a) All 8 KPI tests confirmed arithmetically correct. (b) December appointment type breakdown — Follow-up and Specialist drive the seasonal spike. (c) High Risk best-case rate of 37.93% as a practical intervention benchmark.
Finding from testing	Week 6 features confirmed as analytically valid. Seasonal appointment type breakdown identified as a potential new model feature. 37.93% best-case High Risk rate provides a measurable benchmark.
Action taken	Shared December breakdown by appointment type with Data Science as a candidate time-based feature. Provided 37.93% as the model evaluation benchmark for intervention threshold.
Retest result	All 8 KPI tests PASSED confirming features are analytically correct. Data Science confirmed improvement in model performance using the Week 6 features.
What changed	Data Science adopted lead_time_band and patient_risk_tier as engineered features. 37.93% High Risk best-case rate adopted as the model evaluation benchmark.
Evidence	Formal testing record (Part 5), refinement findings (Part 6), this cross-track record, and the analytical dashboard (Part 7).
How this improves HealthConnect	Validated KPIs and refined segments give the Data Science model a stronger analytical foundation. The 37.93% best-case target gives the entire HealthConnect project a concrete, evidence-backed intervention target.

Part 9 — Assumptions, Limitations, Risks and Dependencies

This section reviews every item identified across Weeks 4 to 6 and documents what has been resolved, what remains relevant, and what is newly discovered or confirmed in Week 7.

Item	Status	Week 7 Update
Date format errors — both columns stored as text	Resolved	Completed in Week 5. All eight Week 7 tests ran without date errors.
90 missing distance values	Resolved	Filled with column mean in Week 5. Distance band analysis validated in Weeks 6 and 7.
Cancelled records handling	Resolved	Excluded from all KPI calculations. Validated correct across Tests T1 to T5.
KPI calculation accuracy	Resolved	All eight formal tests PASSED in Week 7. No arithmetic errors found.
December seasonal driver was unknown	Resolved	Resolved in Week 7 Refinement R1: Follow-up (56.63%) and Specialist (55.10%) are the primary drivers.
Dataset is fictional	Still relevant	All findings remain illustrative. No real clinical decisions should be based solely on this data.
No-show reasons are unknown	Still relevant	Cannot be resolved without additional data collection. A permanent limitation of the dataset.
Seasonal data points are limited (18 months)	Still relevant	December finding is confirmed but needs multiple years of data for statistical confidence.
High Risk best-case target (37.93%) needs clinic validation	New — Week 7	Week 7 quantified 37.93% as the target. Needs real-world clinic testing before implementation.
Final integration and presentation (Week 8)	Pending	Week 8 is final integration and presentation of all project outputs.

Part 10 — Week 7 Project Summary

10.1 What Was Planned

The Week 7 plan was to formally test all five KPI calculations, run analytical consistency checks, investigate three gaps left open by Week 6, validate the cross-track integration with the Data Science track through a test–act–retest cycle, and produce a five-page analytics dashboard.

10.2 What Was Completed

- Eight formal testing tests (T1 to T8) passed with documented expected and actual results.
- Three analytical refinements (R1 to R3) December breakdown, High Risk deep dive, and waiting time analysis.
- Three new charts were produced to support the refinements.
- Cross-track testing cycle completed with the Data Science track.
- Five-page interactive analytics dashboard built and published.
- Formal testing record and refinement record documented.

10.3 Key Validated Findings

- All five Week 5 KPIs are arithmetically correct — no errors found across eight formal tests.

- The December 2025 seasonal spike is driven by Follow-up (56.63%) and Specialist Consultation (55.10%) appointments specifically.
- High Risk patients under best-case conditions (same-week booking + SMS) achieve a 37.93% no-show rate — a 30.31 pp improvement over the same group with no reminder.
- Average patient waiting time is 24.29 minutes. Follow-up appointments have the longest wait (24.77 min). Tuesday is the busiest day (25.34 min).
- SMS remains the best reminder channel across every risk tier without exception.

10.4 Important Decisions Made

37.93% adopted the intervention target: High Risk patients under best-case conditions achieve this rate. It gives the clinic a concrete, evidence-backed improvement target.

Seasonal strategy refined: The December and January intervention should specifically target Follow-up and Specialist Consultation appointments — not all appointment types equally.

Waiting time is treated as experience metric only: It cannot predict no-shows (only attended appointments have a waiting time) but provides valuable insight for patient satisfaction.

10.5 Contribution to the Overall Health Connect Project

All analytical outputs from Weeks 5 and 6 are now formally validated as arithmetically correct. Three refined insights have been added that deepen the clinic's understanding of the December seasonal pattern, High Risk patient behaviour, and the patient waiting experience. The Data Science model has been provided with validated features and a measurable evaluation benchmark. The five-page dashboard makes all findings accessible to clinic management without requiring any Python or SQL knowledge.

10.6 Remaining Work Before Week 8

- Compile all Weeks 5 to 7 findings into final recommendations document.
- Prepare final presentation materials for the HealthConnect project.
- Complete final integration with all project tracks.
- Submit all Week 8 deliverables including the final portfolio presentation.